

Solving Equations with Square Roots

Equations with One Square Root

Solve.

① $\sqrt{x} - 4 = -2$

$$\sqrt{x} = 2$$

$$(\sqrt{x})^2 = 2^2$$

$$\boxed{x = 4}$$

③ $9 = \sqrt{x} + 2$

$$7 = \sqrt{x}$$

$$7^2 = (\sqrt{x})^2$$

$$\boxed{49 = x}$$

⑤ $\frac{\sqrt{z}}{2} + 6 = 9$

$$\sqrt{z} = 6$$

$$(\sqrt{z})^2 = 6^2$$

$$\boxed{z = 36}$$

② $12 = 3\sqrt{y}$

$$\frac{12}{3} = \frac{3\sqrt{y}}{3}$$

$$4 = \sqrt{y}$$

$$4^2 = (\sqrt{y})^2$$

$$\boxed{16 = y}$$

④ $2\sqrt{y} = 4$

$$\sqrt{y} = 2$$

$$(\sqrt{y})^2 = 2^2$$

$$\boxed{y = 4}$$

⑥ $2 = \frac{\sqrt{z}}{3} - 1$

$$9 = \sqrt{z}$$

$$9^2 = (\sqrt{z})^2$$

$$\boxed{81 = z}$$

⑦ $7\sqrt{x} - 1 = 13$

$$\sqrt{x} = 2$$

$$(\sqrt{x})^2 = 2^2$$

$$\boxed{x = 4}$$

⑧ $-10 = \sqrt{x-3} - 4$

$$-6 = \sqrt{x-3}$$

$$(-6)^2 = (\sqrt{x-3})^2$$

$$36 = x-3$$

$$\boxed{39 \neq x} \quad \boxed{\text{No Solution}}$$

⑨ $5 = 7 - 4\sqrt{2y}$

$$-7 + 5 = 7 - 4\sqrt{2y} - 7$$

$$-2 = -4\sqrt{2y}$$

$$\frac{-2}{-4} = \frac{-4\sqrt{2y}}{-4}$$

$$\frac{1}{2} = \sqrt{2y}$$

$$(\frac{1}{2})^2 = (\sqrt{2y})^2$$

$$\frac{1}{4} = 2y$$

$$\frac{1}{4} = \frac{2y}{2}$$

$$\frac{1}{4} \div 2 = y$$

$$\frac{1}{4} \div \frac{2}{1} = y$$

$$\frac{1}{4} \cdot \frac{1}{2} = y$$

$$\boxed{\frac{1}{8} = y}$$

$$(\frac{1}{2})^2 = \frac{1}{2} \rightarrow \frac{1}{2}$$

⑪ $2 = \sqrt{y+1} - 6$

$$8 = \sqrt{y+1}$$

$$8^2 = (\sqrt{y+1})^2$$

$$64 = y+1$$

$$\boxed{63 = y}$$

$$(\frac{4}{3})^2 = \frac{4}{3} \cdot \frac{4}{3} = \frac{16}{9}$$

⑫ $3 = 7 - 3\sqrt{5z}$

$$-7 + 3 = 7 - 3\sqrt{5z} - 7$$

$$-4 = -3\sqrt{5z}$$

$$\frac{-4}{-3} = \frac{-3\sqrt{5z}}{-3}$$

$$\frac{4}{3} = \sqrt{5z}$$

$$(\frac{4}{3})^2 = (\sqrt{5z})^2$$

$$\frac{16}{9} = 5z$$

$$\frac{16}{9} = \frac{5z}{1}$$

$$\frac{16}{9} \div 5 = z$$

$$\frac{16}{9} \cdot \frac{1}{5} = z$$

$$\boxed{\frac{16}{45} = z}$$

$$\begin{aligned} \textcircled{13} \quad 3 &= \sqrt{3-8x} - 2 \\ 5 &= \sqrt{3-8x} \\ 5^2 &= (\sqrt{3-8x})^2 \\ 25 &= 3-8x \\ \frac{22}{-8} &= x \\ \boxed{-\frac{11}{4} = x} \end{aligned}$$

$$\begin{aligned} \textcircled{15} \quad \sqrt{-2+\frac{a}{4}} + 10 &= 12 \\ \sqrt{-2+\frac{a}{4}} &= 2 \\ (\sqrt{-2+\frac{a}{4}})^2 &= 2^2 \\ -2+\frac{a}{4} &= 4 \\ \boxed{a=24} \end{aligned}$$

$$\begin{aligned} \textcircled{17} \quad 8 &= 7 + \frac{\sqrt{2z-4}}{-3} \\ -3 &= \sqrt{2z-4} \\ \boxed{\text{No Solution}} \end{aligned}$$

$$\begin{aligned} \textcircled{14} \quad 1 + \frac{\sqrt{y+7}}{3} &= 2 \\ \sqrt{y+7} &= 3 \\ (\sqrt{y+7})^2 &= 3^2 \\ y+7 &= 9 \\ \boxed{y=2} \end{aligned}$$

$$\begin{aligned} \textcircled{16} \quad 4 &= \sqrt{3-2x} + 2 \\ 2 &= \sqrt{3-2x} \\ 2^2 &= (\sqrt{3-2x})^2 \\ 4 &= 3-2x \\ \boxed{-\frac{1}{2} = x} \end{aligned}$$

$$\begin{aligned} \textcircled{18} \quad 2\sqrt{\frac{x}{4}+5} - 4 &= 6 \\ \sqrt{\frac{x}{4}+5} &= 5 \\ (\sqrt{\frac{x}{4}+5})^2 &= 5^2 \\ \frac{x}{4}+5 &= 25 \\ \boxed{x=80} \end{aligned}$$

$$\begin{aligned} \textcircled{19} \quad 5 + \frac{\sqrt{y+2}}{3} &= 6 \\ \sqrt{y+2} &= 3 \\ (\sqrt{y+2})^2 &= 3^2 \\ y+2 &= 9 \\ \boxed{y=7} \end{aligned}$$

$$\begin{aligned} \textcircled{21} \quad 0 &= \sqrt{2y^2+3y+1} \\ 0^2 &= (\sqrt{2y^2+3y+1})^2 \\ 0 &= 2y^2+3y+1 \\ 0 &= (2y+1)(y+1) \\ \boxed{y=-\frac{1}{2}}, \quad \boxed{y=-1} \end{aligned}$$

Challenge

Multiplying
Two
Binomials

$$\begin{aligned} \textcircled{23} \quad x+4 &= \sqrt{3x+10} \\ (x+4)^2 &= (\sqrt{3x+10})^2 \\ (x+4)(x+4) &= 3x+10 \\ x^2+8x+16 &= 3x+10 \\ x^2+5x+6 &= 0 \\ (x+3)(x+2) &= 0 \\ \boxed{x=-3}, \quad \boxed{x=-2} \end{aligned}$$

$$\begin{aligned} \textcircled{20} \quad 7 &= 9 + \frac{\sqrt{4z-1}}{-2} \\ 4 &= \sqrt{4z-1} \\ 4^2 &= (\sqrt{4z-1})^2 \\ 16 &= 4z-1 \\ \boxed{\frac{17}{4} = z} \end{aligned}$$

$$\begin{aligned} \textcircled{22} \quad \sqrt{y^2-4y-12} &= 0 \\ (\sqrt{y^2-4y-12})^2 &= 0^2 \\ y^2-4y-12 &= 0 \\ (y-6)(y+2) &= 0 \\ \boxed{y=6}, \quad \boxed{y=-2} \end{aligned}$$

$$\begin{aligned} \textcircled{24} \quad x+6 &= \sqrt{5x+26} \\ (x+6)^2 &= (\sqrt{5x+26})^2 \\ (x+6)(x+6) &= 5x+26 \\ x^2+12x+36 &= 5x+26 \\ x^2+7x+10 &= 0 \\ (x+2)(x+5) &= 0 \\ \boxed{x=-2}, \quad \boxed{x=-5} \end{aligned}$$

$$(25) \ y = -1 + \sqrt{9y-11}$$

$$y+1 = \sqrt{9y-11}$$

$$(y+1)^2 = (\sqrt{9y-11})^2$$

$$(y+1)(y+1) = 9y-11$$

$$y^2+2y+1 = 9y-11$$

$$y^2-7y+12=0$$

$$(y-3)(y-4)=0$$

$$\boxed{y=3}, \boxed{y=4}$$

$$(27) \ 7 = \sqrt{56+8z} - z$$

$$7+z = \sqrt{56+8z}$$

$$(7+z)^2 = (\sqrt{56+8z})^2$$

$$(7+z)(7+z) = 56+8z$$

$$49+14z+z^2 = 56+8z$$

$$z^2+6z-7=0$$

$$(z+7)(z-1)=0$$

$$\boxed{z=-7}, \boxed{z=1}$$

$$(26) \ y = \sqrt{5y+6} - 2$$

$$y+2 = \sqrt{5y+6}$$

$$(y+2)^2 = (\sqrt{5y+6})^2$$

$$(y+2)(y+2) = 5y+6$$

$$y^2+4y+4 = 5y+6$$

$$y^2-y-2=0$$

$$(y+1)(y-2)=0$$

$$\boxed{y=-1}, \boxed{y=2}$$

$$(28) \ 5 = \sqrt{7-z} - z$$

$$z+5 = \sqrt{7-z}$$

$$(z+5)^2 = (\sqrt{7-z})^2$$

$$(z+5)(z+5) = 7-z$$

$$z^2+10z+25 = 7-z$$

$$z^2+11z+18=0$$

$$(z+2)(z+9)=0$$

$$\boxed{z=-2}, \boxed{z=-9}$$

Equations with Two Square Roots

$$(29) \ \sqrt{y} = \sqrt{2y-5}$$

$$(\sqrt{y})^2 = (\sqrt{2y-5})^2$$

$$y = 2y-5$$

$$-y = -5$$

$$\boxed{y=5}$$

$$(30) \ \sqrt{y+4} = \sqrt{6y-1}$$

$$(\sqrt{y+4})^2 = (\sqrt{6y-1})^2$$

$$y+4 = 6y-1$$

$$5 = 5y$$

$$\boxed{1=y}$$

$$(31) \ \sqrt{z^2+2} = \sqrt{3z}$$

$$(\sqrt{z^2+2})^2 = (\sqrt{3z})^2$$

$$z^2+2 = 3z$$

$$z^2-3z+2=0$$

$$(z-2)(z-1)=0$$

$$\boxed{z=2}, \boxed{z=1}$$

$$(32) \ \sqrt{z^2-20} = \sqrt{-8z}$$

$$(\sqrt{z^2-20})^2 = (\sqrt{-8z})^2$$

$$z^2-20 = -8z$$

$$z^2+8z-20=0$$

$$(z+10)(z-2)=0$$

$$\boxed{z=-10}, \boxed{z=2}$$

$$\begin{aligned}
 (33) \quad 2\sqrt{x} &= \sqrt{\frac{x}{3}+1} \\
 (2\sqrt{x})^2 &= \left(\sqrt{\frac{x}{3}+1}\right)^2 \\
 (2\sqrt{x})(2\sqrt{x}) &= \frac{x}{3}+1 \\
 4x &= \frac{x}{3}+1 \\
 4x &= \frac{x}{3}+\frac{1}{1} \\
 \frac{4x}{1} &= \frac{x}{3}+\frac{3}{3} \\
 \frac{12x}{3} &= \frac{x+3}{3} \\
 12x &= x+3 \\
 11x &= 3 \\
 \boxed{x = \frac{3}{11}}
 \end{aligned}$$

$$\begin{aligned}
 (34) \quad 3\sqrt{x-2} &= \sqrt{\frac{x}{2}} \\
 (3\sqrt{x-2})^2 &= \left(\sqrt{\frac{x}{2}}\right)^2 \\
 (3\sqrt{x-2})(3\sqrt{x-2}) &= \frac{x}{2} \\
 9(x-2) &= \frac{x}{2} \\
 \frac{9x}{1} - \frac{18}{1} &= \frac{x}{2} \\
 \frac{18x}{2} - \frac{36}{2} &= \frac{x}{2} \\
 \frac{18x-36}{2} &= \frac{x}{2}
 \end{aligned}$$

$18x-36 = x$
 $-36 = -17x$
 $\boxed{\frac{36}{17} = x}$

$$\begin{aligned}
 (35) \quad 5\sqrt{y+4} &= 3\sqrt{2y-9} \\
 (5\sqrt{y+4})^2 &= (3\sqrt{2y-9})^2 \\
 25(y+4) &= 9(2y-9) \\
 25y+100 &= 18y-81 \\
 7y &= -181 \\
 \boxed{y \neq -\frac{181}{7}} \\
 \boxed{\text{No Solution}}
 \end{aligned}$$

$$\begin{aligned}
 (36) \quad 2\sqrt{\frac{y}{2}+4} &= 3\sqrt{3y+1} \\
 (2\sqrt{\frac{y}{2}+4})^2 &= (3\sqrt{3y+1})^2 \\
 4\left(\frac{y}{2}+4\right) &= 9(3y+1) \\
 \frac{4}{1} \cdot \frac{y}{2} + 16 &= 27y+9 \\
 \frac{4y}{2} + 16 &= 27y+9 \\
 2y+16 &= 27y+9 \\
 7 &= 25y \\
 \boxed{\frac{7}{25} = y}
 \end{aligned}$$

$$\textcircled{37} \quad \sqrt{\frac{3z}{4} + 1} = \frac{\sqrt{z+1}}{2}$$

$$\left(\sqrt{\frac{3z}{4} + 1}\right)^2 = \left(\frac{\sqrt{z+1}}{2}\right)^2$$

$$\frac{3z}{4} + 1 = \left(\frac{\sqrt{z+1}}{2}\right)\left(\frac{\sqrt{z+1}}{2}\right)$$

$$\frac{3z}{4} + 1 = \frac{z+1}{4}$$

$$\frac{3z}{4} + \frac{4}{4} = \frac{z+1}{4}$$

$$\frac{3z+4}{4} = \frac{z+1}{4}$$

$$3z+4 = z+1$$

$$2z = -3$$

$$\boxed{z \neq -\frac{3}{2}}$$

No solution

$$\textcircled{38} \quad \sqrt{\frac{5z}{8} - 1} = \frac{\sqrt{z+2}}{4}$$

$$\left(\sqrt{\frac{5z}{8} - 1}\right)^2 = \left(\frac{\sqrt{z+2}}{4}\right)^2$$

$$\frac{5z}{8} - 1 = \left(\frac{\sqrt{z+2}}{4}\right)\left(\frac{\sqrt{z+2}}{4}\right)$$

$$\frac{5z}{8} - 1 = \frac{z+2}{16}$$

$$\frac{10z}{16} - \frac{16}{16} = \frac{z+2}{16}$$

$$\frac{10z-16}{16} = \frac{z+2}{16}$$

$$10z-16 = z+2$$

$$9z = 18$$

$$\boxed{z = 2}$$

challenge

$$\textcircled{39} \quad \sqrt{x} + 3 = \sqrt{x+11}$$

$$(\sqrt{x} + 3)^2 = (\sqrt{x+11})^2$$

$$(\sqrt{x} + 3)(\sqrt{x} + 3) = x + 11$$

$$(\sqrt{x})^2 + 3\sqrt{x} + 3\sqrt{x} + 3^2 = x + 11$$

$$x + 6\sqrt{x} + 9 = x + 11$$

$$-x + x + 6\sqrt{x} + 9 = x + 11 - x$$

$$6\sqrt{x} + 9 = 11$$

$$\sqrt{x} = \frac{2}{6}$$

$$\sqrt{x} = \frac{1}{3}$$

$$(\sqrt{x})^2 = \left(\frac{1}{3}\right)^2$$

$$\boxed{x = \frac{1}{9}}$$

$$\textcircled{40} \quad \sqrt{y+19} = 4 + \sqrt{y}$$

$$(\sqrt{y+19})^2 = (4 + \sqrt{y})^2$$

$$y+19 = (4 + \sqrt{y})(4 + \sqrt{y})$$

$$y+19 = 4^2 + 4\sqrt{y} + 4\sqrt{y} + (\sqrt{y})^2$$

$$y+19 = 16 + 8\sqrt{y} + y$$

$$-y + y + 19 = 16 + 8\sqrt{y} + y - y$$

$$19 = 16 + 8\sqrt{y}$$

$$\frac{3}{8} = \sqrt{y}$$

$$\left(\frac{3}{8}\right)^2 = (\sqrt{y})^2$$

$$\frac{3}{8} \cdot \frac{3}{8} = y$$

$$\boxed{\frac{9}{64} = y}$$

$$\frac{1}{3} \cdot \frac{1}{3} = \frac{1}{9}$$

$$\textcircled{41} \sqrt{y+18} - \sqrt{y+2} = 3$$

$$\sqrt{y+18} = 3 + \sqrt{y+2}$$

$$(\sqrt{y+18})^2 = (3 + \sqrt{y+2})^2$$

$$y+18 = (3 + \sqrt{y+2})(3 + \sqrt{y+2})$$

$$y+18 = 3^2 + 3\sqrt{y+2} + 3\sqrt{y+2} + (\sqrt{y+2})^2$$

$$y+18 = 9 + 6\sqrt{y+2} + y+2$$

$$-y + y + 18 = 9 + 6\sqrt{y+2} + y + 2 - y$$

$$18 = 11 + 6\sqrt{y+2}$$

$$\frac{7}{6} = \sqrt{y+2}$$

$$\left(\frac{7}{6}\right)^2 = (\sqrt{y+2})^2$$

$$\frac{49}{36} = y+2$$

$$\frac{49}{36} - 2 = y$$

$$\frac{49}{36} - \frac{72}{36} = y$$

$$\boxed{-\frac{23}{36} = y}$$

$$\textcircled{42} \sqrt{10+x} = \sqrt{x-1} + 3$$

$$(\sqrt{10+x})^2 = (\sqrt{x-1} + 3)^2$$

$$10+x = (\sqrt{x-1} + 3)(\sqrt{x-1} + 3)$$

$$10+x = (\sqrt{x-1})^2 + 3\sqrt{x-1} + 3\sqrt{x-1} + 3^2$$

$$10+x = x-1 + 6\sqrt{x-1} + 9$$

$$10 = 8 + 6\sqrt{x-1}$$

$$\frac{2}{6} = \sqrt{x-1}$$

$$\frac{1}{3} = \sqrt{x-1}$$

$$\left(\frac{1}{3}\right)^2 = (\sqrt{x-1})^2$$

$$\frac{1}{9} = x-1$$

$$\frac{1}{9} + \frac{1}{1} = x$$

$$\frac{1}{9} + \frac{8}{9} = x$$

$$\boxed{\frac{10}{9} = x}$$

$$\textcircled{43} \sqrt{x-1} = \sqrt{x-8} + 1$$

$$\textcircled{44} \sqrt{x+3} = \sqrt{x+4} - 1$$